

交互题

人员

郭栩睿、陶汇笙、王恩泽 到课

上周作业检查

上周作业链接: <https://cppoj.kids123code.com/contest/4949>

王向东老师周六十点半C++对顶堆						
刷新						
#	用户名	姓名	编程分	时间	A	B
1	taohuisheng	陶汇笙	300	2671	100	100
2	guoxurui	郭栩睿	200	2489	100	100

本周作业

<https://cppoj.kids123code.com/contest/5134> (课上讲了 A ~ D 这些题, 课后作业是 D 题)

课堂表现

今天上课给同学们讲了 交互题 这个知识点, 今天的 D 题比较难, 同学们课上基本都没做出来, 课后要把这道题再做一做。

课堂内容

猜数 (IO交互版)

交互题模板

```
#include <bits/stdc++.h>

using namespace std;

int main()
{
    int l = 1, r = 1e9;
    while (true) {
        int mid = (l + r) / 2;
        cout << mid << endl;

        int x; cin >> x;
        if (x == 0) break;
        if (x == -1) l = mid+1;
        else r = mid-1;
    }
}
```

```
    return 0;
}
```

nextmin

题目允许错一个, 把所有位置问一遍即可

```
#include <bits/stdc++.h>

using namespace std;

const int maxn = 1000 + 5;
int w[maxn];

int main()
{
    int n; cin >> n;
    for (int i = 1; i <= n; ++i) {
        cout << "? " << i << " " << i << endl;
        int x; cin >> x;
        w[i] = x;
    }

    cout << "! ";
    for (int i = 1; i <= n; ++i) cout << w[i] << " ";
    return 0;
}
```

[COI 2020] Zagrade

括号匹配, 每次判断 i 和 栈顶 之间合不合法

```
#include <bits/stdc++.h>

using namespace std;

const int maxn = 1e5 + 5;
char s[maxn];

int main()
{
    int n, Q; cin >> n >> Q;
    stack<int> stk;
    for (int i = 1; i <= n; ++i) {
        if (stk.empty()) stk.push(i);
        else {
            int j = stk.top();
            cout << "? " << j << " " << i << endl;
        }
    }
}
```

```

    int x; cin >> x;
    if (x == 1) s[j] = '(', s[i] = ')', stk.pop();
    else stk.push(i);
}
}

int len = stk.size();
for (int i = 1; i <= len/2; ++i) {
    int x = stk.top(); s[x] = '('; stk.pop();
}
for (int i = 1; i <= len/2; ++i) {
    int x = stk.top(); s[x] = ')'; stk.pop();
}

cout << "! ";
for (int i = 1; i <= n; ++i) cout << s[i];
cout << endl;
return 0;
}

```

[COI 2022] 回文子串 / Madioničar

枚举每个位置作为中心, 往左右枚举长度, 看对应串是不是回文

```

#include <bits/stdc++.h>

using namespace std;

int main()
{
    int n; cin >> n;

    int res = 1;
    for (int i = 1; i <= n; ++i) {
        int len = (res - 1) / 2 + 1;
        while (true) {
            int l = i - len, r = i + len;
            if (l < 1 || r > n) break;

            cout << "? " << l << " " << r << endl;
            int x; cin >> x;
            if (x == 0) break;

            ++len;
        }
        res = 2*(len-1) + 1;
    }

    for (int i = 1; i <= n-1; ++i) {
        int len = (res - 1) / 2 + 1;
        while (true) {

```

```
int l = i - len + 1, r = i + len;
if (l<1 || r>n) break;

cout << "? " << l << " " << r << endl;
int x; cin >> x;
if (x == 0) break;

++len;
}
res = max(res, 2*(len-1));
}

cout << "! " << res << endl;
return 0;
}
```